

■ Learning Check

- The three common phases of matter on Earth are _____, _____, and _____.
- (True or False.) The number of protons in the nucleus of an atom determines which element it is.
- The molecule is the basic unit of all
 - elements.
 - compounds.
 - solutions.
 - mixtures.
- Graphite, diamond, and buckminsterfullerene are different forms of the element _____.
- (True or False.) Air can be compressed easily because there is a great deal of empty space between air molecules.

ANSWERS: 1. solid, liquid, gas 2. True 3. (b) 4. carbon 5. True

■ Learning Check

- If the same-sized force is made to act over a smaller area,
 - the pressure is decreased.
 - the pressure is not changed.
 - the pressure is increased.
 - the result depends on the shape of the area.
- (True or False.) A sealed box contains air under pressure. The outward forces on the walls will be largest on those walls with the smallest area.
- The pressure difference between the inside and the outside of a tire is the _____ pressure.
- (True or False.) If the mass and temperature of the gas in a container stay constant while the volume is decreased, the pressure will also decrease.

ANSWERS: 1. (c) 2. False 3. gauge 4. False

■ Learning Check

- The mass density of an object equals its _____ divided by its _____.
- (True or False.) An alcoholic beverage is a mixture of water and ethyl alcohol. A "stronger" beverage (one with more alcohol in it) is likely to have a higher mass density than a "weaker" one.
- Solid object A weighs more than solid object B. Consequently, we can conclude that
 - A's weight density must be greater than B's.
 - A's volume must be greater than B's.
 - A's weight density and volume must both be greater than B's.
 - Measurements would be required in order to determine the cause of the weight difference.
- The specific gravity of a substance is computed by dividing its density by the density of _____.

ANSWERS: 1. mass, volume 2. False 3. (d) 4. water

■ Learning Check

- (True or False.) Gravity causes the pressure in the ocean to vary with depth.
- Two aquariums are completely full of water, but the pressure at the bottom of aquarium A is greater than at the bottom of aquarium B. The cause of this could be that
 - the water in A is deeper than it is in B.
 - A contains freshwater and B contains seawater.
 - A is shaped like a box and B is shaped like a cylinder.
 - Any of the above.
- (True or False.) The atmospheric pressure at 5,000 m above the ground is exactly twice as large as it is at 10,000 m above the ground.
- A barometer is a device that measures _____.

ANSWERS: 1. True 2. (a) 3. False 4. atmospheric pressure

■ Learning Check

1. The _____ acting on a boat in water at rest is equal to the weight of the water displaced by the boat.
2. One balloon is filled with pure water and a second is filled with antifreeze. What does each balloon do after being released by a scuba diver beneath the surface of the ocean?
3. As a person climbs into a rowboat, it must sink lower into the water so that
 - (a) the net force on the boat remains zero.
 - (b) it displaces more water, consequently increasing the buoyant force on it.
 - (c) its bottom is deeper in the water where the higher pressure causes a larger upward force.
 - (d) All of the above.
4. (True or False.) The buoyant force on a hydrogen-filled balloon is larger than that on a helium-filled balloon with the same size.

ANSWERS: 1. buoyant force 2. pure water balloon rises, the other sinks 3. (d) 4. False

■ Learning Check

1. Squeezing the sides of a flexible water bottle to make water come out of the top is an example of exploiting _____ principle.
2. (True or False.) When the pressure in the fluid in two cylinders in a hydraulic system is the same, the forces on the two pistons are not necessarily equal.

ANSWERS: 1. Pascal's 2. True

■ Learning Check

1. Bernoulli's principle is exploited by devices such as _____.
2. (True or False.) At a place in a particular ventilation duct where the air pressure is lowest, the air is likely to be moving fastest.

ANSWERS: 1. perfume atomizer or carburetor 2. True